



Kingdom of Saudi Arabia
Ministry of Education
King Faisal University
CCSIT, King Faisal University

College of Computer Sciences & Information Technology

AI Based Research Documents Repository Using Local Language Models

*A project submitted
in partial fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science*

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September 2026

Acknowledgment

(Replace the text below with your own acknowledgment.)

We would like to express our sincere gratitude to our supervisor, **Nora Abdullah Alkhaldi**, for the continuous guidance, valuable feedback, and support throughout the development of this graduation project. We also extend our appreciation to the committee members and all faculty and staff in College of Computer Sciences & Information Technology for their assistance and encouragement.

Undertaking

This is to declare that the Graduation Project entitled “**AI Based Research Documents Repository Using Local Language Models**” is an original work carried out by the undersigned, in partial fulfillment of the requirements for the degree **Bachelor of Science in Computer Science** at the Department of CS, College of Computer Sciences & Information Technology, King Faisal University.

All the analysis, design, implementation, and system development activities have been accomplished by the undersigned. Furthermore, this project has not been submitted, in whole or in part, to any other college or university for any degree or qualification.

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Abstract

[Write your abstract here (250–300 words).]

Keywords: keyword1; keyword2; keyword3; keyword4

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List of Acronyms

API Application Programming Interface

AI Artificial Intelligence

NLP Natural Language Processing

GPU Graphics Processing Unit

1 First-Heading

1.1 First-Sub-heading-1

1.1.1 First-Sub-subheading-1-a

1.1.2 First-Sub-subheading-1-b

1.2 First-Sub-heading-2

1.2.1 First-Sub-subheading-2-a

1.2.2 First-Sub-subheading-2-b

2 Second-Heading

2.1 Second-Sub-heading-1

2.1.1 Second-Sub-subheading-1-a

2.1.2 Second-Sub-subheading-1-b

2.2 Second-Sub-heading-2

2.2.1 Second-Sub-subheading-2-a

2.2.2 Second-Sub-subheading-2-b

This project follows software engineering principles [1]. Similar AI-based solutions have been reported in recent studies [2]. A comparable system design was presented in [3].

Acronyms are expanded automatically on first use. The proposed system exposes an Application Programming Interface ([API](#)) that applies Artificial Intelligence ([AI](#)) and Natural Language Processing ([NLP](#)) techniques on a Graphics Processing Unit ([GPU](#)), and every later mention of the [API](#) prints the short form only.

Bibliography

- [1] R. Tanaka, T. Iki, T. Hasegawa, K. Nishida, K. Saito, and J. Suzuki, “Vdocrag: Retrieval-augmented generation over visually-rich documents,” *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. doi: [10.48550/arXiv.2504.09795](https://doi.org/10.48550/arXiv.2504.09795)
- [2] K.-C. Dong et al., “Doc-researcher: A unified system for multimodal document parsing and deep research,” *Proceedings of the ACM Web Conference 2026*, 2026. doi: [10.1145/3774904.3792599](https://doi.org/10.1145/3774904.3792599)
- [3] D. Gosmar and G. Zenezini, “Madp: A multi-agent pipeline for sustainable document processing with human-in-the-loop,” *ArXiv*, 2026. doi: [10.48550/arXiv.2605.17159](https://doi.org/10.48550/arXiv.2605.17159)

Appendix A: Additional Materials

The following equation is provided only as an illustrative example to demonstrate how to include a general numbered equation in this report template.

$$y = ax + b \quad (1)$$

$$y = ax + b \quad (2a)$$

$$x = \frac{y - b}{a} \quad (2b)$$

$$x^{10} = y - b \quad (2c)$$

$$f^{(k)}(x) \neq f_i(x) \quad (2d)$$

Table 1 is included only as an example to demonstrate table formatting in this report template.

Table 1: General comparison between existing solutions and the proposed system.

Criteria	Existing Solutions	Proposed System
Ease of use	Moderate	High
Accuracy of results	Moderate	Improved
User interface clarity	Basic	Enhanced
Automation level	Limited	High
Reliability	Moderate	High
Scalability	Limited	Good
Reporting and documentation	Basic	Detailed

The following table is provided only as an illustrative example to demonstrate how to include a bordered table with **merged column headers** in this report template (Table 2).

Table 2: Comparison of system characteristics.

Feature	System Type	
	Traditional Approach	Proposed Approach
User experience	Average	Improved
System flexibility	Low	High
Maintenance effort	High	Lower
Processing speed	Moderate	Faster
Overall efficiency	Moderate	High

The following table is provided only as an illustrative example to demonstrate how to include a bordered table with **merged rows** in this report template (Table 3).

Table 3: General comparison of project features.

Category	Criteria	Existing	Proposed
Usability	Ease of use	Medium	High
	Accessibility	Medium	High
Performance	Reliability	Medium	High
	Response speed	Medium	High
Output Quality	Report clarity	Basic	Detailed
	Output consistency	Medium	High

The following table is provided only as an illustrative example to demonstrate how to include a bordered table with **merged rows and merged column headers** in this report template (Table 4).

Table 4: Illustrative comparison table (example only).

Category	Criteria	Comparison	
		Option A	Option B
Usability	Ease of use	High	Medium
	Clarity of interface	High	Medium
Quality	Reliability	High	Medium
	Output clarity	High	Medium
Efficiency	Time required	Low	Medium
	Effort required	Low	Medium

The following code listing is provided only as an illustrative example to demonstrate how to include a numbered and captioned code excerpt in this report template (Listing 1).

Listing 1: Example function that normalizes an input string.

```

1 def normalize(text):
2     """Return the text in lower case, without leading or trailing spaces."""
3     return text.strip().lower()

```

As shown in Listing 1, source code is typeset with line numbers and a caption placed above the excerpt. Set language to the language of your own code, for example Java, C, Python, or SQL. Include only short, meaningful excerpts in the report; complete source files belong in the submitted code archive.

Appendix B: Project Screenshots



Figure 1: Official CCSIT logo used in this graduation project report.

As shown in Figure 1, the official CCSIT logo is presented for institutional identification and report branding.



(a) King Faisal University (KFU) logo.



(b) College of Computer Sciences & Information Technology (CCSIT) logo.

Figure 2: Institutional logos used for official report branding.

Figure 2 presents the institutional logos used in this report.